Higher National Diploma in Engineering (Electrical)
4th Year, First Semester Examination – 2022
EE3215 - Broadband Communication

Instructions for Candidates:
Answer any five 05 questions
All Questions carry equal Marks

No. of questions: 06
No. of pages : 04
Time : 3 Hours

01.

i. Write a short answers.

(a.) Explain, Broadband communication.

(b.) Why use broadband communication?

(c.) The main advantage of a broadband communication system over narrow-band communication.

(d.) Which technique is most popular in the transmission of digital data?

(e.) What is the main advantage of using a broadband transmission medium?

(f.) What is the difference between satellite and wireless broadband communication systems?

(02x 6=12 Marks)

ii. Fill the table given below. **(08 Marks)**

DSL Type	Description	Up/Down Symmetry	Data Rate Downstream	Data Rate Upstream	Distance Limit	Application
ADSL						
SDSL						
HDSL						
VDSL						

(Total 20 Marks)

02.

Generally, a broadband communication system provides channels for the transmission of data in several directions by many users. For instance, the broadband communication system transmits several TV channels over a similar cable and also handles data transmissions within a completely different frequency spectrum. In the competition among broadband service providers, cable has outsold its competitors by about 2:1, with a telephone technology known as *Digital Subscriber Line* (DSL) providing the strongest competition to date.

i. Which types of cables are used for cable television? **(02 Marks)**

ii. Explain DSL? **(03 Marks)**

iii. What is the difference between cable and DSL broadband communication systems? **(03 Marks)**

iv. What are the advantages of DSL broadband over other broadband technologies? **(03 Marks)**

v. What are the transmission constraints on DSL due to Copper wire and how they affect the DSL signal transmission? **(05 Marks)**

vi. What is the impact of loading coils on a DSL? **(04 Marks)**

(Total 20 Marks)

03.

Asymmetric digital subscriber line (ADSL) is a type of digital subscriber line (DSL) technology, a data communications technology that enables faster data transmission over copper telephone lines than a conventional voice band modem can provide. ADSL differs from the less common symmetric digital subscriber line (SDSL).

i. Draw the connectivity diagram of ADSL connection showing different blocks from the Central Office to Subscriber Premises. **(6 Marks)**

ii. Why is ADSL more popular than SDSL? Give both technical and marketing reasons. **(6 Marks)**

iii. What are the two types of modulation techniques used by xDSL Technologies? Give a brief description for each illustrating the spectrum of each, modulated signal. **(8 Marks)**

(Total 20 Marks)

04.

A large range of communication technologies with different technical capacities are capable of providing high-speed internet to households. Wired technologies include copper cable (xDSL), coaxial cable (HFC), broadband over power lines (BPL), and optical fiber cable (FTTx).

Copper wires are defined as 'legacy telephone unshielded copper twisted pair', providing broadband connections by using xDSL technologies.

i. Write pros and cons for use of copper wires for the communication. **(04 Marks)**

ii. What is broadband over power line (BPL)? Briefly explain. **(04 Marks)**

iii. What are the pros and cons of BPL? **(04 Marks)**

iv. Discussed use of optical fibers for communication. **(04 Marks)**

v. What are the pros and cons of optical fibers for xDSL technologies? **(04 Marks)**

(Total 20 Marks)

05.

In the world of high-speed data communications, and Internet access in particular, services offered by cable operators have co-developed with other wired and wireless options. For residential users, Internet access was historically provided almost exclusively through dial-up modems, directly to *Internet Service Providers* (ISPs).

i. What is the internet? **(03 Marks)**

ii. What meant by ISP? **(03 Marks)**

iii. Explain Asynchronous Transfer Mode (ATM) and how it supports data communication. **(05 Marks)**

iv. Write description about DSLAM effect to high-speed data communication. **(05 Marks)**

v. Explain difference between DSL and DSLAM. **(04 Marks)**

(Total 20 Marks)

06.

Wireless broadband technologies include mobile radio solutions (e.g., HSPA, LTE), fixed radio solutions (e.g. WiMAX), and satellite solutions.

i. Write short notes WiMAX, HSPA and LTE. **(09 Marks)**

ii. Explain the effect satellite solutions on wireless broadband with pros and cons. **(04 Marks)**

iii. What is Light Fidelity (Li-Fi) and how does it differ from Wi-Fi? **(04 Marks)**

iv. How do Internet balloons support wireless broadband? **(03 Marks)**

(Total 20 Marks)